

Homework for week 1, submission deadline 23:59 on Sunday night.

1. Complete an Omics-to-Question map: Using your own research project or biological interest. What kind of omics you plan to use to solve the biological research question?

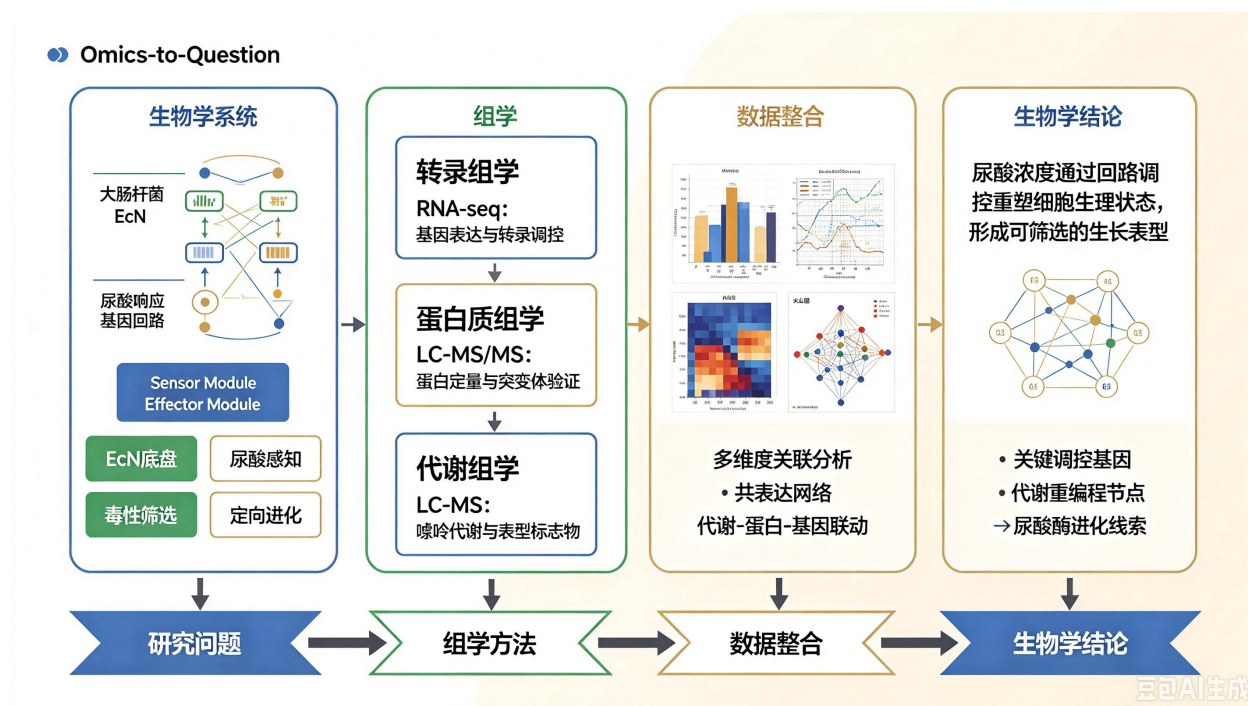
biological research question: 如何通过生长偶联筛选 (Growth-Coupled Selection) 策略, 定向进化出在肠道厌氧微环境中具有更高催化活性和氧利用效率的尿酸酶 (Uricase) 变体, 从而构建一种基于工程益生菌 (Escherichia coli Nissle 1917, EcN) 的、无需抗生素筛选标记的安全高效高尿酸血症活体生物治疗平台?

Whole-Genome Sequencing, WGS

RNA-seq

Label-free Quantitative Proteomics

Targeted Metabolomics



2. Write one weak prompt and one CARE+V prompt: For the same biological or research question, propose a weak and strong (CARE+V) prompt that can be tested by AI. Use any LLMs to test your prompt and paste the output results below.

Please help me analyze the growth curve data from my uricase directed evolution experiment.

Result:

Based on the provided slides, here is an analysis of the growth curve data from your uricase directed evolution experiment.

Summary of Experimental Logic

The core of your experiment relies on a toxin-mediated, growth-coupled screening system. This is detailed in image 8.png and image 9.png.

Sensor Module (Module A): A constitutive promoter drives the expression of the HucR repressor and the YgfU transporter. YgfU imports uric acid (UA) from the environment into the cell.

Effector/Selection Module (Module B): The HucR repressor binds to the PHucR promoter, blocking the expression of a toxic protein, SUMO-GD-N

The Mechanism of Action:

In the presence of Uric Acid (UA), the imported UA binds to the HucR repressor, causing it to detach from the PHucR promoter.

This derepresses the PHucR promoter, leading to the expression of the SUMO-GD-N toxin.

The production of this toxin inhibits cell growth, resulting in a lower optical density (OD600)

Therefore, there is an inverse relationship: Higher Uric Acid concentration $\rightarrow$ Higher Toxin expression $\rightarrow$ Lower Cell Growth (OD600).

Analysis of Growth Response Data

The growth response data is presented in image 11.png through image 14.png. The graph in these images plots the Maximum OD600 (y-axis) against the Uric Acid Concentration (μ M) (x-axis).

1. Comparison of Strains (Treatment Groups)

SUMO-GD-N (Red Triangles): This is your test strain, containing the functional uric acid-responsive toxin circuit.

This strain shows the strongest growth inhibition in response to uric acid.

As UA concentration increases from 0 μ M to 1000 μ M, the OD600 drops sharply, from approximately 1.4 to below 1.0.

This confirms that the sensor module is successfully detecting uric acid and activating the toxin.

SUMO-GD-N-4A (Blue Circles): This is the negative control strain, where the toxin (SUMO-GD-N) has been inactivated via four mutations (R137A/K145A/R151A/R153A).

This strain shows significantly less growth inhibition compared to the active SUMO-GD-N strain.

While there is a slight decrease in OD600 at higher UA concentrations (likely due to metabolic burden or non-specific toxicity of the protein), the slope is much less steep than the active toxin strain. This demonstrates that the growth inhibition observed in the test strain is specifically due to the active toxin.

Empty Vector (Green Triangles / "empty"): This is the baseline control, lacking the toxin gene entirely.

This strain maintains the highest overall OD600 across the uric acid gradient.

The slight decrease at high UA concentrations (e.g., 1000 μM) likely represents the basal toxicity or stress of the bacterium importing and processing a high amount of uric acid. The fact that the SUMO-GD-N-4A strain behaves similarly to the empty vector control further validates the negative control.

2. Statistical Analysis Summary

The statistical analyses (ANOVA and Tukey's post-hoc test) confirm that the differences observed are statistically significant.

Table 2 (Linear Regression Parameters):

The SUMO-GD-N strain shows the strongest negative correlation, with an R^2 of 0.8810 and a steep negative slope of -0.0004882.

The SUMO-GD-N-4A control has a much smaller slope (-0.0001842), and the Empty Vector has the smallest slope (-0.0001707), indicating minimal growth response to UA.

Table 3 (Two-Way ANOVA):

Both Uric Acid concentration and Treatment group are highly significant sources of variation in the data ($P < 0.0001$).

The significant Interaction term ($P = 0.0002$) confirms that the effect of uric acid concentration on growth is dependent on the specific strain used, which is exactly what your hypothesis predicts.

Table 4 (Tukey's Pairwise Comparison):

Comparing SUMO-GD-N vs. SUMO-GD-N-4A: The mean difference in OD600 is 0.08261, with the difference being statistically significant (***, $P = 0.0007$).

Comparing SUMO-GD-N vs. Empty Vector: The mean difference is -0.1398, with very high significance ($P < 0.0001$).

These post-hoc tests confirm that the test strain's growth is significantly different from both controls.

Conclusion

The growth curve data fully validates the design and function of your genetic circuit. The toxin-mediated growth-coupled screening logic is confirmed: an increase in uric acid concentration leads to a significant and specific

decrease in cell growth for the strain carrying the active toxin circuit, while control strains remain relatively unaffected.

This establishes a robust platform for your next steps in uricase directed evolution. As you evolve variants of uricase, the "best" mutants (those that degrade UA most efficiently) will lower the intracellular concentration of uric acid. This will lead to less toxin expression and, consequently, higher cell growth compared to less efficient variants at the same initial uric acid concentration. This will allow you to screen for improved uricase activity simply by selecting for faster-growing colonies.

Context: I am a synthetic biology researcher working on directed evolution of uricase using a growth-coupled selection system in *Escherichia coli* Nissle 1917 (EcN). I have performed a kinetic microplate assay with four experimental groups (GDN test, 4A negative control, GFP positive control, Empty baseline) across six uric acid concentrations (0, 50, 100, 200, 500, 1000 μM). OD600 and GFP fluorescence were measured every 5 minutes for 15 hours at 37°C. Each condition has three biological replicates ($n=3$). The goal is to validate that the uric acid-responsive circuit successfully couples uric acid concentration to growth inhibition via the GD-N toxin, enabling growth-coupled screening.

Ask: Please write a complete R script that:

Imports and cleans the microplate kinetic data.

Calculates the mean and standard error of the mean (SEM) for OD600 across replicates for each time point, group, and uric acid concentration

Generates two publication-quality figures: (a) a faceted growth curve (OD600 vs. time) for all four groups at each uric acid concentration, and (b) a dose-response bar plot showing OD600 at 15 hours for each group across all uric acid concentrations.

Performs a two-way ANOVA to test the main effects of uric acid concentration, group, and their interaction on growth phenotype.

Rules:

Use only the `tidyverse`, `ggplot2`, and `ggpubr` packages.

All plots must use the journal-style theme (minimal, black axis lines, no gridlines).

Figures must be saved as high-resolution PDFs (300 dpi).

Code must include comprehensive inline comments explaining each step.

Error bars must represent SEM.

Examples: Please follow the plotting style and statistical presentation format commonly found in *Nature Microbiology* or *ACS Synthetic Biology* growth curve figures (faceted layout with clear group legends).

Verification:

Ensure the script checks that all required packages are installed before running.
Include a data integrity check that warns if any replicates are missing.
Add a `sessionInfo()` output at the end of the script for reproducibility.
The code should handle both OD600 and GFP data for potential future analysis.

Result:

An internal error has occurred.

```
1 #-----
2 # R Script: Analysis for Uric-Acid Responsive Growth-Coupled Selection Circuit
3 # Project: Directed Evolution of Uricase in Escherichia coli Nissle 1917 (EcN)
4 # Assay: Kinetic microplate assay, OD600 + GFP, n=3 biological replicates
5 # Packages: tidyverse, ggplot2, ggpubr only
6 # Output: 2 publication-grade PDF figures, two-way ANOVA, data integrity checks
7 #-----
8
9 # ----- Step 1: Check & Load Required Packages -----
10 required_pkgs <- c("tidyverse", "ggplot2", "ggpubr")
11 new_pkgs <- required_pkgs[!(required_pkgs %in% installed.packages()[,"Package"])]
12 if(length(new_pkgs) > 0){
13   install.packages(new_pkgs, dependencies = TRUE)
14 }
15
16 # Load Libraries
17 library(tidyverse)
18 library(ggplot2)
19 library(ggpubr)
20
21 # ----- Step 2: Import & Clean Microplate Kinetic Data -----
22 ## Expected input CSV columns:
23 # Time_h: numeric, time in hours (0-15)
```

```

24 # Group: categorical, one of c("GDN", "4A", "GFP", "Empty")
25 # UA_uM: numeric uric-acid concentration (0,50,100,200,500,1000)
26 # Replicate: integer biological replicate ID (1,2,3)
27 # OD600: optical density reading
28 # GFP: raw fluorescence reading (kept for future downstream use)
29
30 # Import raw data; replace file path with your actual data location
31 raw_data <- read_csv("microplate_kinetic_data.csv", show_col_types = FALSE)
32
33 # Data integrity check 1: expected factor levels
34 expected_groups <- c("GDN", "4A", "GFP", "Empty")
35 expected_ua <- c(0,50,100,200,500,1000)
36 expected_replicates <- c(1,2,3)
37
38 # Check for missing replicates per (Group, UA_uM) combination
39 complete_combinations <- expand.grid(
40   Group = expected_groups,
41   UA_uM = expected_ua,
42   Replicate = expected_replicates
43 )
44
45 missing_records <- complete_combinations %>%
46   anti_join(raw_data, by = c("Group", "UA_uM", "Replicate"))
47
48 if(nrow(missing_records) > 0){
49   warning("WARNING: Missing biological replicate records detected!")
50   print(missing_records)
51 } else {
52   message("Data integrity check passed: No missing replicate combinations found.")
53 }

```

```

54
55 # Clean & type-cast dataset
56 clean_data <- raw_data %>%
57   mutate(
58     Group = factor(Group, levels = expected_groups),
59     UA_uM = factor(UA_uM, levels = expected_ua),
60     Replicate = as.integer(Replicate)
61   ) %>%
62   drop_na(Time_h, Group, UA_uM, Replicate, OD600) # drop rows with critical missing values
63
64 # ----- Step3: Compute Summary Statistics (Mean + SEM) -----
65 # SEM = standard error of the mean = sd / sqrt(n)
66 kinetic_summary <- clean_data %>%
67   group_by(Time_h, Group, UA_uM) %>%
68   summarise(
69     OD_mean = mean(OD600, na.rm = TRUE),
70     OD_sd = sd(OD600, na.rm = TRUE),
71     n = n(),
72     OD_sem = OD_sd / sqrt(n),
73     GFP_mean = mean(GFP, na.rm = TRUE),
74     GFP_sem = sd(GFP, na.rm = TRUE)/sqrt(n),
75     .groups = "drop"
76   )
77
78 # Extract endpoint data (15-hour time point) for dose-response plot
79 endpoint_data <- kinetic_summary %>%
80   filter(near(Time_h, 15, tol = 0.1)) # match 15h with small tolerance for timestamp rounding
81
82 # Raw endpoint data for statistical testing (use raw replicate values, not summary stats for ANOVA)
83 endpoint_raw <- clean_data %>%

```

```

84   filter(near(Time_h,15, tol=0.1))
85
86   # ----- Step4: Publication-Quality Theme (Nature Microbiology / ACS Synth Biol style) -----
87   journal_theme <- theme_minimal() +
88     theme(
89       axis.line = element_line(color = "black", linewidth = 0.4),
90       panel.grid = element_blank(),
91       axis.text = element_text(color = "black", size = 9),
92       axis.title = element_text(color = "black", size = 10),
93       strip.text = element_text(size=9, face = "plain"),
94       legend.position = "bottom",
95       legend.title = element_text(size=9),
96       legend.text = element_text(size=8),
97       plot.margin = margin(4,4,4,4),
98       panel.spacing = unit(0.4, "cm")
99     )
100
101   # ----- Figure A: Faceted Growth Curves (OD600 vs Time) per UA concentration -----
102   growth_curve_plot <- ggplot(kinetic_summary, aes(x = Time_h, y = OD_mean, color = Group)) +
103     geom_line(linewidth = 0.7) +
104     geom_errorbar(aes(ymin = OD_mean - OD_sem, ymax = OD_mean + OD_sem), width = 0.15, alpha =0.6) +
105     facet_wrap(~UA_uM, labeller = labeller(UA_uM = ~paste0(.x, "  $\mu$ M UA")) +
106     labs(
107       x = "Time (h)",
108       y = expression(OD[600]),
109       color = "Experimental Group"
110     ) +
111     scale_color_brewer(palette = "Set2") +
112     journal_theme

```

```

113
114 ggsave(
115   filename = "FigureA_growth_curves.pdf",
116   plot = growth_curve_plot,
117   width = 10,
118   height = 7,
119   dpi = 300,
120   device = "pdf"
121 )
122
123 message("Saved Figure A: faceted growth curves as FigureA_growth_curves.pdf")
124
125 # ----- Figure B: Dose-Response Bar Plot at 15h Endpoint -----
126 dose_response_plot <- ggplot(endpoint_data, aes(x = UA_uM, y = OD_mean, fill = Group)) +
127   geom_col(position = position_dodge(width = 0.8), width = 0.7) +
128   geom_errorbar(
129     aes(ymin = OD_mean - OD_sem, ymax = OD_mean + OD_sem),
130     position = position_dodge(width = 0.8),
131     width = 0.2
132   ) +
133   labs(
134     x = "Uric Acid Concentration ( $\mu$ M)",
135     y = expression(OD[600] @ 15 h),
136     fill = "Experimental Group"
137   ) +
138   scale_fill_brewer(palette = "Set2") +
139   journal_theme
140
141 ggsave(
142   filename = "FigureB_dose_response_15h.pdf",

```

```

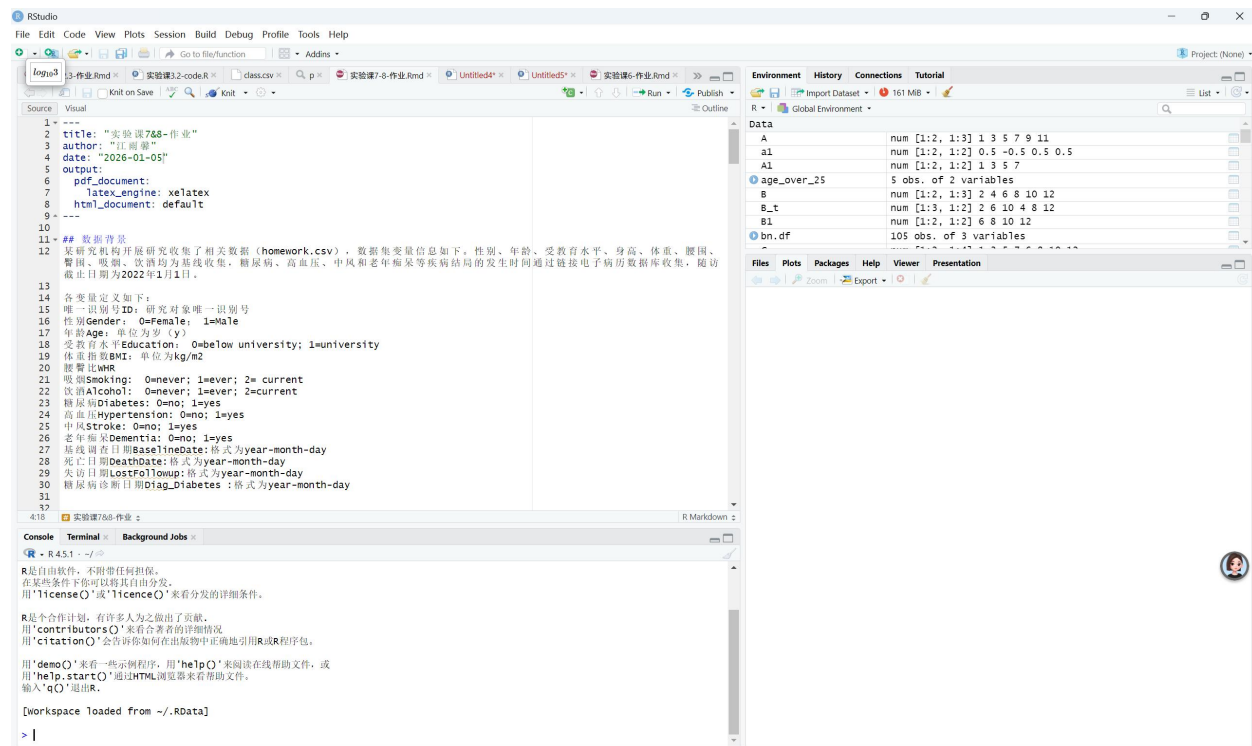
143 plot = dose_response_plot,
144 width =9,
145 height =6,
146 dpi =300,
147 device = "pdf"
148 )
149
150 message("Saved Figure B: dose-response bar plot as FigureB_dose_response_15h.pdf")
151
152 # ----- Two-Way ANOVA: test UA concentration, Group, interaction effect -----
153 # Use RAW replicate endpoint values, not aggregated means for valid statistical inference
154 endpoint_raw <- endpoint_raw %>%
155   mutate(
156     UA_uM = as.factor(UA_uM),
157     Group = as.factor(Group)
158   )
159
160 two_way_anova <- aov(OD600 ~ UA_uM * Group, data = endpoint_raw)
161 anova_summary <- summary(two_way_anova)
162
163 cat("\n===== Two-Way ANOVA Output =====\n")
164 print(anova_summary)
165
166 # Optional: Tukey post-hoc pairwise comparison
167 tukey_result <- TukeyHSD(two_way_anova)
168 cat("\n===== Tukey HSD Post-hoc Pairwise Comparisons =====\n")
169 print(tukey_result)
170
171 # ----- Reproducibility: print session information -----
172 cat("\n===== Session Info for Reproducibility =====\n")

```

3. Install and Run R and RStudio and save output showing R version and packages.

R: <https://cran.r-project.org/> RStudio: <https://posit.co/products/open-source/rstudio>.

Once installation, please paste a screen cut of Rstudio here. Give a try of R to using the folder (Week1_R_RStudio_Practice_Files) code and data.



```
> R.version.string[1] "R version 4.5.1 (2025-06-13 ucrt)"
```

Rows: 12 Columns: 5

— column specification —

Delimiter: `","` **chr** (4): sample_id, group, sex, batch **dbl** (1): body_weight_g
 i Use ``spec()`` to retrieve the full column specification for this data. i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.
`counts <- read_csv("toy_gene_counts.csv")` Rows: 8 Columns: 13

— column specification —

Delimiter: `","` **chr** (1): gene_id **dbl** (12): s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12
 i Use ``spec()`` to retrieve the full column specification for this data. i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.
`enzyme <- read_csv("enzyme_activity.csv")` Rows: 12 Columns: 4

— column specification —

```

Delimiter: ","chr (2): sample_id,
groupdbl (2): citrate_synthase_activity, lactate_mmol_L
i Use `spec()` to retrieve the full column specification for this data. i Spe
cify the column types or set `show_col_types = FALSE` to quiet this message.>
# Inspect the first rows> head(metadata)# A tibble: 6 x 5
  sample_id group sex batch body_weight_g
  <chr> <chr> <chr> <chr> <dbl>
1 S1 Control F B1 22.12
2 S2 Control M B1 21.84
3 S3 Control F B3 22.76
4 S4 Control M B2 24.53
5 S5 Control M B3 25.25
6 S6 Control M B3 24.9

# Inspect the first rows> head(counts)# A tibble: 6 x 13
  gene_id S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 S11
  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 Nr4a1 101 140 129 124 133 119 244 229 243 218 2
2 Thbs4 78 72 82 78 75 93 127 116 106 146
3 Myod1 200 223 198 177 210 190 262 286 269
4 Pax7 141 151 156 151 134 162 113 126 1
5 Il6 64 64 59 78 56 51 94 101
6 Gapdh 989 970 992 931 965 941 973
1015 1057 1003 1055 1025

```

4. Install agent and create your first prompt log entry documenting AI use and verification.

4.1 Github: <https://www.github.com/> (Create your account and repository for your homework submission). You need an account and a repository for homework submission (following the pattern: Bioinformatics_homework_Yourname_ID). You also need personal access token in together with repository link to activate your EasyMultiProfiler-web account.

4.2 Hermes-agent: <https://github.com/NousResearch/hermes-agent> (Install agent for coding) and Hermes WebUI (<https://github.com/nesquena/hermes-webui>)

4.3 Codex: <https://chatgpt.com/codex/> (Install for coding and agent)

4.4 Deepseek API: <https://platform.deepseek.com/usage> (Top up ¥10 for agent set and switching to LLM accessed only on SUAT campus) or NVIDIA API

HERMES-AGENT



deepseek-v4-flash · Nous Research
C:\Users\13751\AppData\Local\hermes\hermes-agent
Session: 20260830_212553_85844d

Hermes Agent v0.20.6 (2026.8.27) · upstream 66666f6e

Available Tools
browser: browser_back, browser_click, ...
browser-use: browser_exec
clarify: clarify
code_execution: execute_code
computer_use: computer_use
cronjob: cronjob
delegation: delegate_task
file: patch, read_file, search_files, write_file
(and 19 more toolsets...)

Available Skills
autonomous-ai-agents: claude-code, codex, computer-use, hermes-agent, opencode
creative: architecture-diagram, ascii-video, baoyu-infographic, claude-design, design-md, +5 more
devops: sdhc-review
email: email-inbox-triage, himalaya
media: gif-search, songsee, youtube-content
note-taking: obsidian
productivity: airtable, box, document-to-action-items, docx, google-workspace, maps, +8 more
research: arxiv, competitor-news-monitor, grounded-citations, llm-wiki
software-development: codebase-inspection, dogfood, github, hermes-agent-skill-authoring, +7 more
web: blocked-page-recovery

22 tools · 52 skills · /help for commands

Welcome to Hermes Agent! Type your message or /help for commands.

+ Tip: hermes curator rollback restores skills from a pre-run snapshot - backups live under skills/curator_backups/.

● 请帮我写一个R脚本，用于读取CSV格式的微孔板数据，并绘制不同时间点的生长曲线。
Initializing agent...

```
^ preparing write_file.  
■ preparing terminal.  
^ write C:\Users\13751\growth_curve.R 4.2s  
review diff  
a/C:/Users/13751\growth_curve.R → b/C:/Users/13751\growth_curve.R  
@@ -0,0 +1,171 @@  
##  
## /usr/bin/env Rscript  
##  
## 微孔板生长曲线绘图脚本 (Growth Curve Plotter for Microplate Data)
```



部署完毕了但是似乎有点崩溃

```
# preparing terminal...
$ python -c " d = open(r'C:/Users/13751/growth_curve.R', encoding='ascii').read() open('crLf_test.R', 'w', encoding='ascii').write(d) # python text mode -> CRLF on Windows print('crLf bytes:', len(open('crLf_test.R', 'rb').read())) * + 1 command 6.2s
```

Hermes

CRLF 版完美运行！测 LF 转换后的 ascii_test.R 验证行尾假设：

```
# preparing terminal...
$ python -c " d = open('ascii_test.R', encoding='ascii').read() open('ascii_lf.R', 'w', encoding='ascii', newline='\n').write(d) print('LF bytes:', len(open('ascii_lf.R', 'rb').read())) * + 2 commands 3.5s
```

Hermes

ascii_lf.R (6581 字节 LF) 也成功一行尾不是主因。做内容二分：去掉长头部注释，保留全部代码再测：

```
# preparing terminal...
$ python -c " d = open(r'C:/Users/13751/growth_curve.R', encoding='utf-8').read() lines = d.split('\n') idx = next(i for i,l in enumerate(lines) if 'command line arguments' in l) out = ['#!usr/bin/env Rscript', '# short header'] + lines[idx:] open('t1_nohdr.R', 'w', encoding='ascii', newline='\n').write('\n'.join(out)) print('wrote', len('\n'.join(out)), 'bytes') * + 1 command 4.5s
```

Hermes

无长注释头 (5181 字节) 完全正常！问题在头部注释块。二定位具体注释行：

```
# preparing terminal...
$ python -c " d = open(r'C:/Users/13751/growth_curve.R', encoding='utf-8').read() lines = d.split('\n') idx = next(i for i,l in enumerate(lines) if 'command line arguments' in l) hdr = lines[1:idx] # header lines (excl shebang, excl code) print('header lines:', len(hdr)) half = len(hdr)//2 for name, part in [(('t2a', hdr[half]), ('t2b', hdr[half])), ('t2c', part)]: n = max(0, (target - base) // len(pad)) out = d + pad * n open(f'sz_{target}.R', 'w', encoding='ascii', newline='\n').write(part) print(name, 'bytes:', len('\n'.join(out))) * + 4 commands 5.6s
```

Hermes

两半都正常，但合在一起前——这是纯字节数相关的问题！填充注释行验证大小阈值：

```
# preparing terminal...
$ python -c " d = open('t1_nohdr.R', encoding='utf-8').read() base = len(d.encode('ascii')) pad = '#' padding comment line for size testing 8123456789\n' for target in [6800, 6800, 7200, 7600, 7600]: n = max(0, (target - base) // len(pad)) out = d + pad * n open(f'sz_{target}.R', 'w', encoding='ascii', newline='\n').write(out) print(f'sz-{target}', len(out.encode('ascii'))) * + 3 commands 10.1s
```

Hermes

填充到 7578 字节全正常——不是大小，是特定内容布局。唯一特殊的是 \uffff 转义。测试去掉它：

```
# preparing terminal...
$ python -c " d = open(r'C:/Users/13751/growth_curve.R', encoding='utf-8').read() d2 = d.replace('lines[1]' <- sub('""\\uffff', '\\"', lines[1])', 'lines[1]' <- sub('""x"', '\\"', lines[1])) print("replaced:", d2 != d) open('t3_nofeff.R', 'w', encoding='ascii', newline='\n').write(d2) print('bytes:', len(d2.encode('ascii'))) * + 1 command 4.3s
```

Hermes

找到了！问题是脚本里的 \uffff 转义序列—R 4.5.1 的 Rscript 在 Windows 上解析它时崩溃（崩溃位置随其在文件中的偏移变化）。用 intToUtf8(0xFF) 在运行时构造该字符，源码里不再含 \u 转义：

```
# preparing patch...
$ patch C:\Users\13751\growth_curve.R 2.6s
review diff
a/C:/Users/13751\growth_curve.R > b/C:/Users/13751\growth_curve.R
@@ -70,7 +70,10 @@
    00 - 70.7 + 70.10.00
```

5. Install EasyMultiProfiler and EasyMultiProfiler-Web for multi-omics data analysis and homework submission (You can use Hermes or Hermes WebUI to complete the installation step)
EasyMultiProfiler: <https://github.com/liubingdong/EasyMultiProfiler>

EasyMultiProfiler-Web: <https://github.com/xielab2017/EasyMultiProfiler-Web>

Homework submission

1. Create your github account
2. Create your repository for this course. The naming style: Bioinformatics_homework_Yourname_ID
3. Scan QR code below and submit your infor



Bioinformatics_homework_XieLiwei Public

1 main 1 Branch 0 Tags

Go to file T Add file + Code

File	Commit	Time
xielab2017 [EMP] sync Week_12/multomics/weekly (CHIPseq 联合分析) v8.0.0 run 202...	c70bc13	last month
EMP2026 [EMP] sync Week_12/multomics/weekly (CHIPseq 联合分析)		last month
.gitattributes	Submit Homework_1: literature reference document	last month
.gitignore	Submit Homework_1: literature reference document	last month
README.md	Update course details in README.md	last month

README

Bioinformatics_homework_XieLiwei

Shenzhen University of Advanced Technology, Faculty of Synthetic Biology Course Name: Bioinformatics: From Multi-omics to Discovery Github Homework submission system

About

SUAT_Bioinformatics_2026_Fall

Readme

Activity

Stars

Watching

Forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

xielab2017 Xie Lab